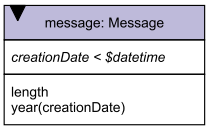
## BI / read / 1

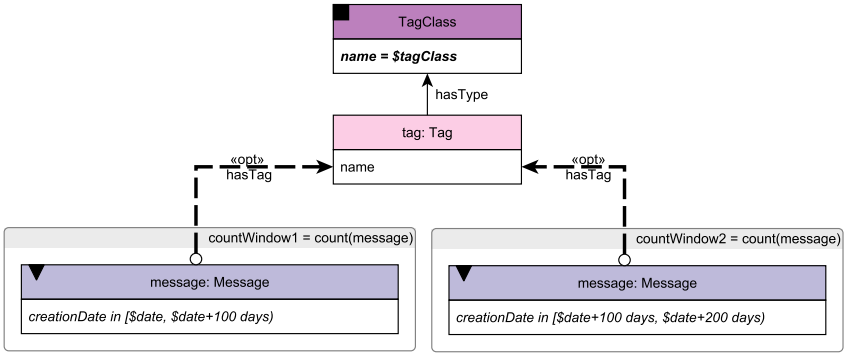
BI 1
BI 2
BI 3
BI 4
BI 5
BI 6
BI 7
BI 8
BI 9
BI 10
BI 11
BI 12
BI 13
BI 14
BI 15
BI 16
BI 17
BI 18
BI 19
BI 20

| | |
|---|---|
| query | BI / read / 1 |
| title | Posting summary |
| pattern |  |
| description | Given a `$datetime`, find all Messages created before that moment. Group them by a 3-level grouping:<br>1. by year of creation<br>2. for each year, group into Message types: is Comment or not<br>3. for each year-type group, split into four groups based on length of their content<br>• `0`: $0 \leq$ length $< 40$ (short)<br>• `1`: $40 \leq$ length $< 80$ (one liner)<br>• `2`: $80 \leq$ length $< 160$ (tweet)<br>• `3`: $160 \leq$ length (long) |
| params | 1 `$datetime` DateTime |
| result | |
| sort | |
| limit | n/a |
| CPs | 1.2, 3.2, 4.1, 4.2, 8.5 |

| | | | | |
|---|---|---|---|---|
| 1 | `year` | 32-bit Integer | R | `year(message.creationDate)` |
| 2 | `isComment` | Boolean | M | `True` for Comments, `False` for Posts |
| 3 | `lengthCategory` | 32-bit Integer | C | `0` for short, `1` for one-liner, `2` for tweet, `3` for long |
| 4 | `messageCount` | 64-bit Integer | A | Total number of Messages in that group |
| 5 | `averageMessageLength` | 32-bit Float | A | Average length of the Message content in that group |
| 6 | `sumMessageLength` | 64-bit Integer | A | Sum of all Message content lengths |
| 7 | `percentageOfMessages` | 32-bit Float | A | Number of Messages in group as a percentage of all messages created before the given date |

| | | | |
|---|---|---|---|
| 1 | `year` | ↓ | |
| 2 | `isComment` | ↑ | `False` < `True`, i.e. Posts come first and Comments second |
| 3 | `lengthCategory` | ↑ | |

## BI / read / 2

| | |
|---|---|
| query | BI / read / 2 |
| title | Tag evolution |
| pattern |   |
| description | Find the Tags under a given `$tagClass` that were used in Messages during in the 100-day time window starting at `$date` and compare it with the 100-day time window that follows. For the Tags and for both time windows, compute the count of Messages. |
| params | |

| | | | |
|---|---|---|---|
| 1 | `$date` | Date | Based on the creation day – TagClass – number of Messages factor table:<br>`(a)` A flashmob date<br>`(b)` A non-flashmob date |
| 2 | `$tagClass` | Long String | For both `(a)` and `(b)`, TagClasses with a similar amount of Messages are selected |

result

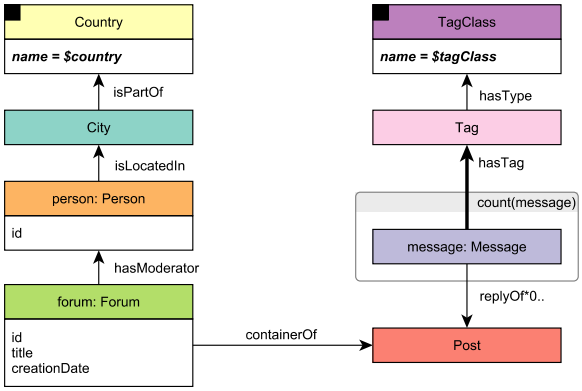
| | | | | |
|---|---|---|---|---|
| 1 | `tag.name` | Long String | R | |
| 2 | `countWindow1` | 32-bit Integer | A | Occurrences of the `tag` during the first time window |
| 3 | `countWindow2` | 32-bit Integer | A | Occurrences of the `tag` during the second time window |
| 4 | `diff` | 32-bit Integer | A | Absolute difference of `countWindow1` and `countWindow2` |

sort

| | | | |
|---|---|---|---|
| 1 | `diff` | ↓ | |
| 2 | `tag.name` | ↑ | |

| | |
|---|---|
| limit | 100 |
| CPs | 2.4, 3.1, 3.2, 4.1, 4.2, 4.3, 5.3, 6.1, 8.2, 8.5 |

## BI / read / 3

| | |
|---|---|
| query | BI / read / 3 |
| title | Popular topics in a country |
| pattern |   |
| description | Given a `$tagClass` and a `$country`, find all the Forums created in the given `$country`, containing at least one Message with Tags belonging directly to the given `$tagClass`, and count the Messages by the Forum which contains them.<br>The location of a Forum is identified by the location of the Forum's moderator. |

**params**

| | | | |
|---|---|---|---|
| 1 | `$tagClass` | Long String | TagClasses with a similar amount of Messages are selected |
| 2 | `$country` | Long String | Big Countries are selected |

**result**

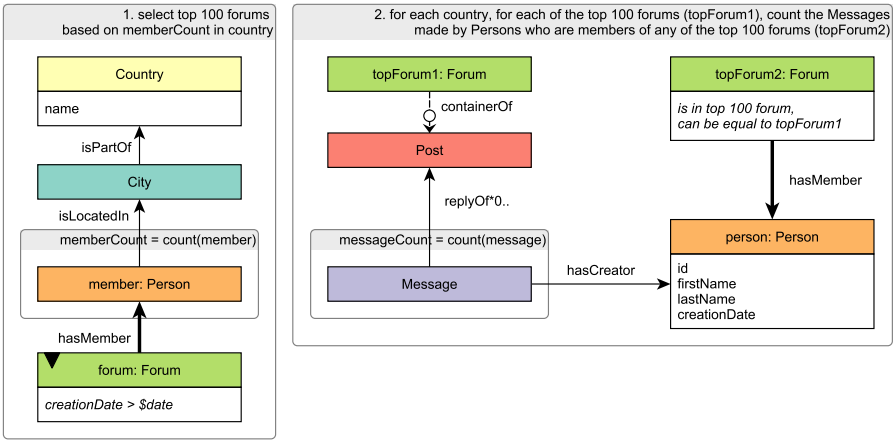
| | | | | |
|---|---|---|---|---|
| 1 | `forum.id` | ID | R | |
| 2 | `forum.title` | Long String | R | |
| 3 | `forum.creationDate` | DateTime | R | |
| 4 | `person.id` | ID | R | |
| 5 | `messageCount` | 32-bit Integer | A | |

**sort**

| | | | |
|---|---|---|---|
| 1 | `messageCount` | ↓ | |
| 2 | `forum.id` | ↑ | |

| | |
|---|---|
| limit | 20 |
| CPs | 1.1, 1.2, 1.3, 2.1, 2.2, 2.4, 3.3, 8.2 |

## BI / read / 4

| | |
|---|---|
| query | BI / read / 4 |
| title | Top message creators by country |
| pattern |  |
| description | Find the most popular Forums by Country, where the popularity of a Forum is measured by the number of members that Forum has from a given Country and the Forum was created after a given `$date`.<br>Calculate the top 100 most popular Forums. If a Forum is popular in multiple countries, it should only be calculated once with its largest membership. In case of a tie, the Forum with the smaller id value should be selected.<br>For each member Person of the 100 most popular Forums, count the number of Messages (`messageCount`) they made in any of those (most popular) Forums. Also include those member Persons who have not posted any Messages (have a `messageCount` of 0). |
| params | |
| result | |
| sort | |
| limit | 100 |
| CPs | 1.2, 1.3, 2.1, 2.2, 2.3, 2.4, 3.3, 5.3, 6.1, 8.2, 8.4 |



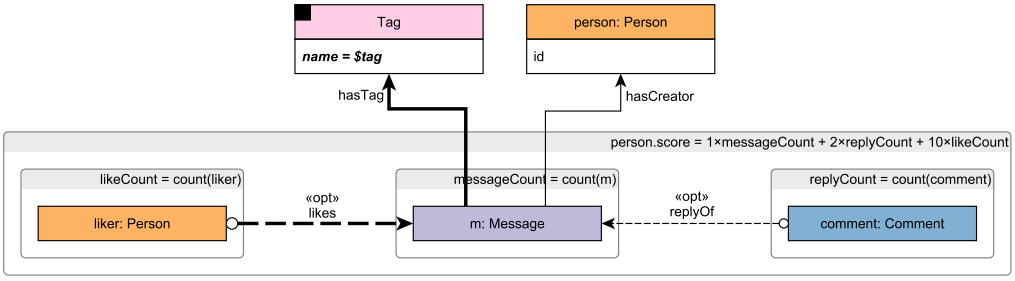
| | | | |
|---|---|---|---|
| 1 | `$date` | Date | Selected from the first 30 days of the network |

| | | | | |
|---|---|---|---|---|
| 1 | `person.id` | ID | R | |
| 2 | `person.firstName` | String | R | |
| 3 | `person.lastName` | String | R | |
| 4 | `person.creationDate` | DateTime | R | |
| 5 | `messageCount` | 32-bit Integer | A | |

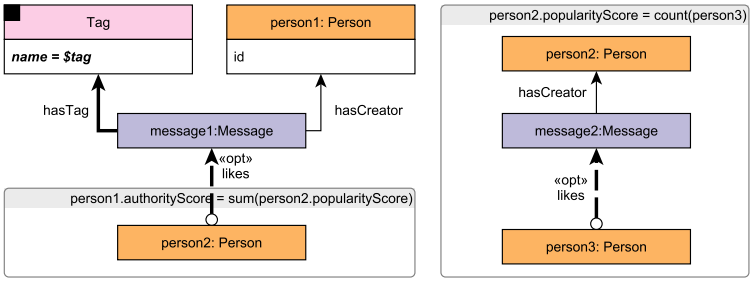
| | | | |
|---|---|---|---|
| 1 | `messageCount` | ↓ | |
| 2 | `person.id` | ↑ | |

## BI / read / 5

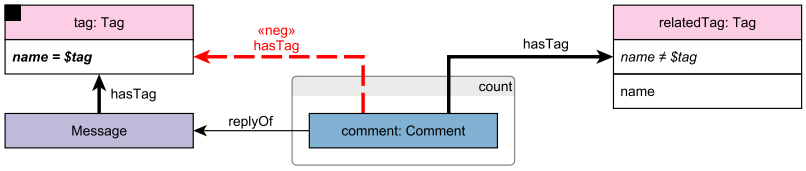
BI 1
BI 2
BI 3
BI 4
BI 5
BI 6
BI 7
BI 8
BI 9
BI 10
BI 11
BI 12
BI 13
BI 14
BI 15
BI 16
BI 17
BI 18
BI 19
BI 20

| | |
|---|---|
| query | BI / read / 5 |
| title | Most active posters of a given topic |
| pattern |  |
| description | Get each Person (`person`) who has created a Message (`message`) with a given `$tag` (direct relation, not transitive). Considering only these Messages, for each Person node:<br>• Count its Messages (`messageCount`).<br>• Count likes (`likeCount`) to its Messages.<br>• Count Comments (`replyCount`) in reply to its Messages.<br>The `score` is calculated according to the following formula: $1 \times$ `messageCount` $+ 2 \times$ `replyCount` $+ 10 \times$ `likeCount`. |
| params | 1 \| `$tag` \| Long String \| Tags with a similar amount of Messages are selected. To avoid caching, different Tags should be used than the ones in Q6 and Q7. |
| result | 1 \| `person.id` \| ID \| R<br>2 \| `replyCount` \| 32-bit Integer \| A<br>3 \| `likeCount` \| 32-bit Integer \| A<br>4 \| `messageCount` \| 32-bit Integer \| A<br>5 \| `score` \| 32-bit Integer \| A |
| sort | 1 \| `score` \| ↓<br>2 \| `person.id` \| ↑ |
| limit | 100 |
| CPs | 1.2, 2.3, 2.6, 8.2 |

## BI / read / 6

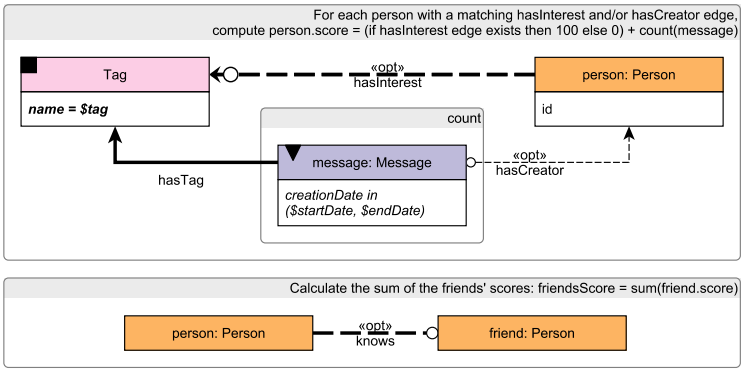
| | |
|---|---|
| query | BI / read / 6 |
| title | Most authoritative users on a given topic |
| pattern |  |
| description | Given a `$tag`, find all Persons (`person1`) that ever created a Message with the `$tag`. For each of these Persons (`person1`) compute their "authority score" as follows:<br>• The "authority score" is the sum of "popularity scores" of the Persons (`person2`) that liked any of that Person's Messages with the given `$tag` (same criterion as for `message1`).<br>• A Person's (`person2`) "popularity score" is defined as the total number of likes (by any Person `person3`) on any of their Messages (`message2`). |
| params | 1 \| `$tag` \| Long String \| Tags with a similar amount of Messages are selected. To avoid caching, different Tags should be used than the ones in Q5 and Q7. |
| result | 1 \| `person1.id` \| ID \| R<br>2 \| `authorityScore` \| 32-bit Integer \| A |
| sort | 1 \| `authorityScore` \| ↓<br>2 \| `person1.id` \| ↑ |
| limit | 100 |
| CPs | 1.2, 2.3, 2.6, 3.3, 6.1, 8.2 |
| relevance | Computing the authority scores might involve computing the popularity score for the same Person multiple times. Implementations are advised to avoid such redundant computations. |

## BI / read / 7

| | |
|---|---|
| query | BI / read / 7 |
| title | Related topics |
| pattern |  |
| description | Find all Messages that have a given `$tag`. Find the related Tags attached to (direct) reply Comments of these Messages, but only of those reply Comments that do not have the given `$tag`.<br>Group the related Tags by name, and get the count of replies in each group. |
| params | 1 \| `$tag` \| Long String \| Tags with a similar amount of Messages are selected. To avoid caching, different Tags should be used than the ones in Q5 and Q6. |
| result | 1 \| `relatedTag.name` \| Long String \| R<br>2 \| `count` \| 32-bit Integer \| A |
| sort | 1 \| `count` \| ↓<br>2 \| `relatedTag.name` \| ↑ |
| limit | 100 |
| CPs | 1.4, 3.3, 5.2, 8.1 |

## BI / read / 8

BI 1
BI 2
BI 3
BI 4
BI 5
BI 6
BI 7
BI 8
BI 9
BI 10
BI 11
BI 12
BI 13
BI 14
BI 15
BI 16
BI 17
BI 18
BI 19
BI 20

| | |
|---|---|
| query | BI / read / 8 |
| title | Central person for a tag |
| pattern |  |
| description | Given a `$tag`, find all Persons that are interested in the `$tag` and/or have written a Message (Post or Comment) with a `creationDate` after a given `$startDate` and that has a given `$tag`. For each Person, compute the `score` as the sum of the following two aspects:<br>• 100, if the Person has this `$tag` as their interest, or 0 otherwise<br>• number of Messages by this Person with the given `$tag`<br>Also, for each Person, compute the sum of the score of the Person's friends (`friendsScore`). |

params

| | | | |
|---|---|---|---|
| 1 | `$tag` | Long String | Tags with a similar amount of Messages are selected |
| 2 | `$startDate` | Date | `(a)`: A range during which a flashmob event happened (it should yield at least a 5× difference)<br>`(b)`: A regular range (does not include a flashmob event) |
| 3 | `$endDate` | Date | |

result

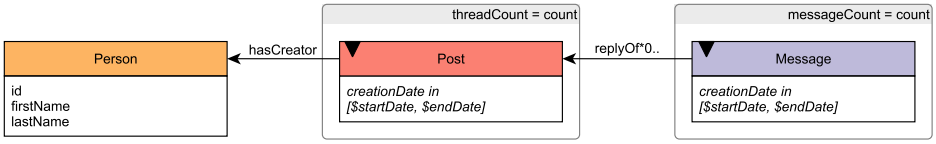
| | | | | |
|---|---|---|---|---|
| 1 | `person.id` | ID | R | |
| 2 | `score` | 32-bit Integer | A | |
| 3 | `friendsScore` | 32-bit Integer | A | The sum of the score of the `person`'s friends |

sort

| | | | |
|---|---|---|---|
| 1 | `score + friendsScore` | ↓ | |
| 2 | `person.id` | ↑ | |

| | |
|---|---|
| limit | 100 |
| CPs | 1.2, 2.1, 2.3, 3.2, 5.3, 8.2, 8.4, 8.5 |
| relevance | Similarly to BI 16, there are two major ways to compute this query: (1) creating an induced subgraph of the interested Persons and their friends and performing the scoring on this graph or (2) performing the scoring without creating an induced subgraph and scoring the friends of a Person on-the-fly. The first approach is more efficient as it avoids redundant computations, however, specifying it needs support for composable graph queries. |

## BI / read / 9

| | |
|---|---|
| query | BI / read / 9 |
| title | Top thread initiators |
| pattern |  |
| description | For each Person, count the number of Posts they created in the time interval `[$startDate, $endDate]` (equivalent to the number of threads they initiated) and the number of Messages in each of their (transitive) reply trees, including the root Post of each tree. When calculating Message counts only consider Messages created within the given time interval.<br>Return each Person, number of Posts they created, and the count of all Messages that appeared in the reply trees (including the Post at the root of tree). |
| params | |

| | | | |
|---|---|---|---|
| 1 | `$startDate` | Date | Selected around the same date |
| 2 | `$endDate` | Date | 80-100 days after the `$startDate` |

result

| | | | | |
|---|---|---|---|---|
| 1 | `person.id` | ID | R | |
| 2 | `person.firstName` | String | R | |
| 3 | `person.lastName` | String | R | |
| 4 | `threadCount` | 32-bit Integer | A | The number of Posts created by that Person (the number of threads initiated) |
| 5 | `messageCount` | 32-bit Integer | A | The number of Messages created in all the threads this Person initiated |

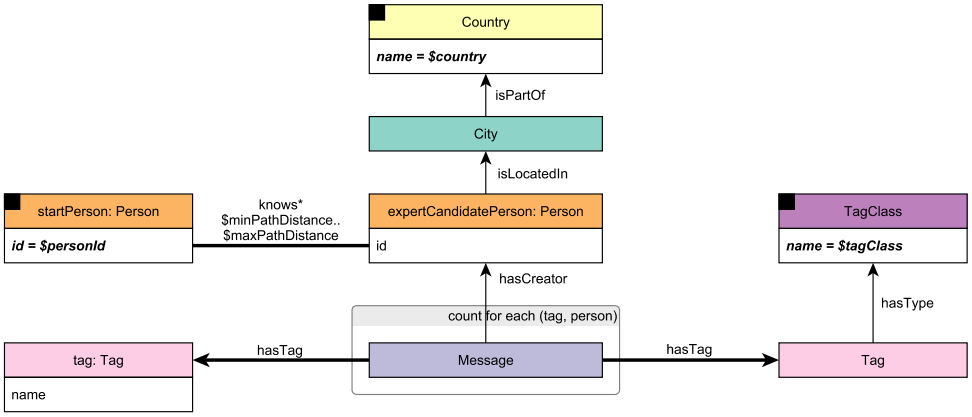
sort

| | | | |
|---|---|---|---|
| 1 | `messageCount` | ↓ | |
| 2 | `person.id` | ↑ | |

| | |
|---|---|
| limit | 100 |
| CPs | 1.2, 2.2, 2.3, 2.6, 3.2, 7.2, 7.3, 7.4, 8.1, 8.5 |

## BI / read / 10

BI 1
BI 2
BI 3
BI 4
BI 5
BI 6
BI 7
BI 8
BI 9
BI 10
BI 11
BI 12
BI 13
BI 14
BI 15
BI 16
BI 17
BI 18
BI 19
BI 20

| | |
|---|---|
| query | BI / read / 10 |
| title | Experts in social circle |
| pattern |  |
| description | Given a Person `startPerson` with ID `$personID`, find all other Persons (`expertCandidatePerson`) that live in a given `$country` and are connected to the `startPerson` on a *shortest path* with length in range `[$minPathDistance, $maxPathDistance]` through the knows relation.<br>For each of these `expertCandidatePerson` nodes, retrieve all of their Messages that contain at least one Tag belonging to a given `$tagClass` (direct relation not transitive). For each Message, retrieve all of its Tags.<br>Group the results by Persons and Tags, then count the Messages by a certain Person having a certain Tag. |

**params**

| | | | |
|---|---|---|---|
| 1 | `$personId` | ID | (a) Persons with an average degree of knows edges are selected<br>(b) Persons who have only one friend and that Person has two friends in total (including the original Person) |
| 2 | `$country` | String | Select mid-sized Countries |
| 3 | `$tagClass` | Long String | TagClasses with a similar degree of hasType edges are selected |
| 4 | `$minPathDistance` | 32-bit Integer | 3 |
| 5 | `$maxPathDistance` | 32-bit Integer | 4 |

**result**

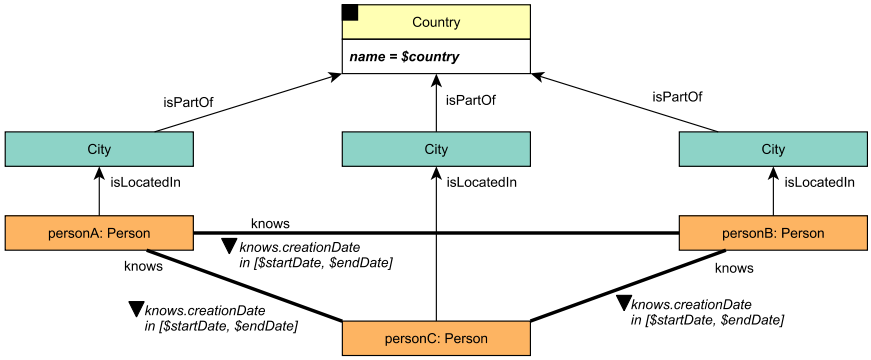
| | | | | |
|---|---|---|---|---|
| 1 | `expertCandidatePerson.id` | ID | R | |
| 2 | `tag.name` | Long String | R | |
| 3 | `messageCount` | 32-bit Integer | A | Number of Messages created by that Person containing that Tag |

**sort**

| | | | |
|---|---|---|---|
| 1 | `messageCount` | ↓ | |
| 2 | `tag.name` | ↑ | |
| 3 | `expertCandidatePerson.id` | ↑ | |

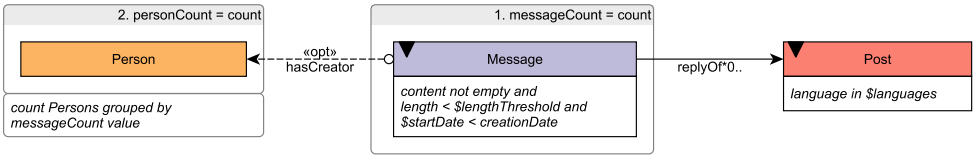
| | |
|---|---|
| limit | 100 |
| CPs | 1.2, 1.3, 2.3, 2.4, 2.6, 3.3, 5.3, 7.1, 7.2, 7.3, 8.1, 8.6 |

## BI / read / 11

| | |
|---|---|
| query | BI / read / 11 |
| title | Friend triangles |
| pattern |   |
| description | For a given `$country`, count all the distinct triples of Persons such that:<br>• `personA` is friend of `personB`,<br>• `personB` is friend of `personC`,<br>• `personC` is friend of `personA`,<br>and these friendships were created in the range `[$startDate, $endDate]`.<br>Distinct means that given a triple $t_1$ in the result set $R$ of all qualified triples, there is no triple $t_2$ in $R$ such that $t_1$ and $t_2$ have the same set of elements. |
| params | 1 `$country` Long String Selected from the largest Countries (India, China)<br>2 `$startDate` Date Selected from a 30-day interval towards the end of the simulation time<br>3 `$endDate` Date Selected to yield around a 100-day interval |
| result | 1 count 64-bit Integer A |
| limit | n/a |
| CPs | 2.3, 2.5, 3.2 |

## BI / read / 12

BI 1
BI 2
BI 3
BI 4
BI 5
BI 6
BI 7
BI 8
BI 9
BI 10
BI 11
BI 12
BI 13
BI 14
BI 15
BI 16
BI 17
BI 18
BI 19
BI 20

| | |
|---|---|
| query | BI / read / 12 |
| title | How many persons have a given number of messages |
| pattern |  |
| description | For each Person, count the number of Messages they made (`messageCount`). Only count Messages with the following attributes:<br>• Its `content` is not empty (and consequently, the `imageFile` attribute is empty for Posts).<br>• Its `creationDate` is after `$startDate` (exclusive, equality is not allowed).<br>• Its `length` is below the `$lengthThreshold` (exclusive, equality is not allowed).<br>• It is written in any of the given `$languages`.<br>– The language of a Post is defined by its `language` attribute.<br>– The language of a Comment is that of the Post that initiates the thread where the Comment replies to.<br>The Post and Comments in the reply tree's path (from the Message to the Post) do not have to satisfy the constraints for `content`, `length`, and `creationDate`.<br>For each `messageCount` value, count the number of Persons with exactly `messageCount` Messages (with the required attributes). |

**params**

| | | | |
|---|---|---|---|
| 1 | `$startDate` | Date | Selected randomly from a 60-day interval. |
| 2 | `$lengthThreshold` | 32-bit Integer | Balanced against `startDate` to filter around 30% of the Messages within a language and keep the variance low.<br>The selection of this parameter uses a factor table of bucketed Message lengths and creation dates. |
| 3 | `$languages` | {String} | Only the most frequently used languages |

**result**

| | | | | |
|---|---|---|---|---|
| 1 | `messageCount` | 32-bit Integer | A | Number of Messages created |
| 2 | `personCount` | 32-bit Integer | A | Number of Persons with `messageCount` Messages |

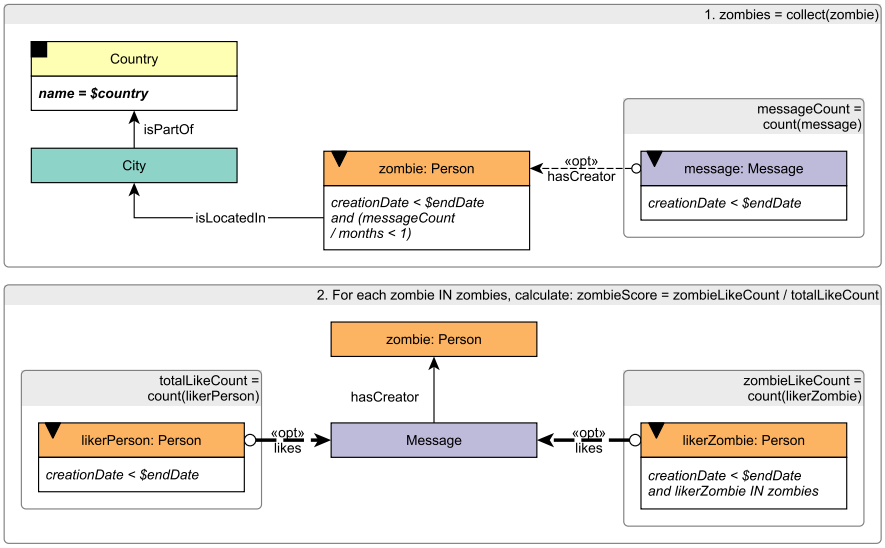
**sort**

| | | |
|---|---|---|
| 1 | `personCount` | ↓ |
| 2 | `messageCount` | ↓ |

| | |
|---|---|
| limit | n/a |
| CPs | 1.1, 1.2, 1.4, 2.6, 3.2, 4.2, 4.3, 8.1, 8.2, 8.3, 8.4, 8.5 |

## BI / read / 13

BI 1
BI 2
BI 3
BI 4
BI 5
BI 6
BI 7
BI 8
BI 9
BI 10
BI 11
BI 12
BI 13
BI 14
BI 15
BI 16
BI 17
BI 18
BI 19
BI 20

| | |
|---|---|
| query | BI / read / 13 |
| title | Zombies in a country |
| pattern |  |
| description | Find zombies within the given `$country`, and return their zombie scores. A `zombie` is a Person created before the given `$endDate`, which has created an average of `[0, 1)` Messages per month, during the time range between profile's `creationDate` and the given `$endDate`. The number of months spans the time range from the `creationDate` of the profile to the `$endDate` with partial months on both end counting as one month (e.g. a `creationDate` of Jan 31 and an `$endDate` of Mar 1 result in 3 months).<br>For each `zombie`, calculate the following:<br>• `zombieLikeCount`: the number of likes received from other zombies.<br>• `totalLikeCount`: the total number of likes received.<br>• `zombieScore`: `zombieLikeCount` / `totalLikeCount`. If the value of `totalLikeCount` is 0, the `zombieScore` of the `zombie` should be 0.0.<br>For both `zombieLikeCount` and `totalLikeCount`, only consider likes received from profiles that were created before the given `$endDate`. |

**params**

| | | | |
|---|---|---|---|
| 1 | $country | Long String | Selected from the largest Countries (India, China) |
| 2 | $endDate | Date | Selected from the last days of the initial data set |

**result**

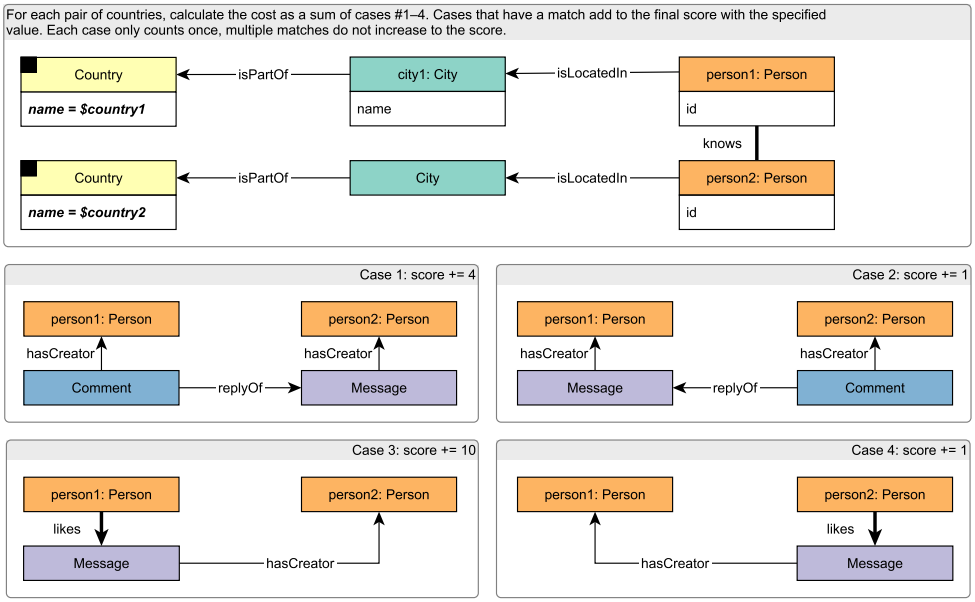
| | | | | |
|---|---|---|---|---|
| 1 | zombie.id | ID | R | |
| 2 | zombieLikeCount | 32-bit Integer | A | |
| 3 | totalLikeCount | 32-bit Integer | A | |
| 4 | zombieScore | 32-bit Float | A | Determined as `zombieLikeCount` / `totalLikeCount` |

**sort**

| | | | |
|---|---|---|---|
| 1 | zombieScore | ↓ | |
| 2 | zombie.id | ↑ | |

| | |
|---|---|
| limit | 100 |
| CPs | 1.2, 2.1, 2.3, 2.4, 2.6, 3.2, 3.3, 4.2, 5.1, 5.3, 8.2, 8.4, 8.5 |

## BI / read / 14

| | |
|---|---|
| query | BI / read / 14 |
| title | International dialog |
| pattern |  |
| description | Consider all pairs of people (`person1`, `person2`) such that (1) they know each other, (2) one is located in a City of `$country1`, and (3) the other is located in a City of `$country2`. For each City of `$country1`, return the highest scoring pair. If there are multiple top-scoring pairs in a city, return the pair with the lowest (`person1.id`, `person2.id`) using a lexicographical ordering.<br>The score of a pair is defined as the sum of the subscores awarded for the following kinds of interaction. The initial value is `score = 0`.<br>1. `person1` has created a reply Comment to at least one Message by `person2`: `score += 4`<br>2. `person1` has created at least one Message that `person2` has created a reply to: `score += 1`<br>3. `person1` liked at least one Message by `person2`: `score += 10`<br>4. `person1` has created at least one Message that was liked by `person2`: `score += 1`<br>Consequently, the maximum score a pair can obtain is: `4 + 1 + 10 + 1 = 16`. |



params

| | | | |
|---|---|---|---|
| 1 | $country1 | Long String | (a) Correlated with parameter `country2`, i.e. the Countries are close and there are many Persons knowing each other<br>(b) Uncorrelated with parameter `country2`, i.e. the Countries are afar and there are few Persons knowing each other |
| 2 | $country2 | Long String | |

result

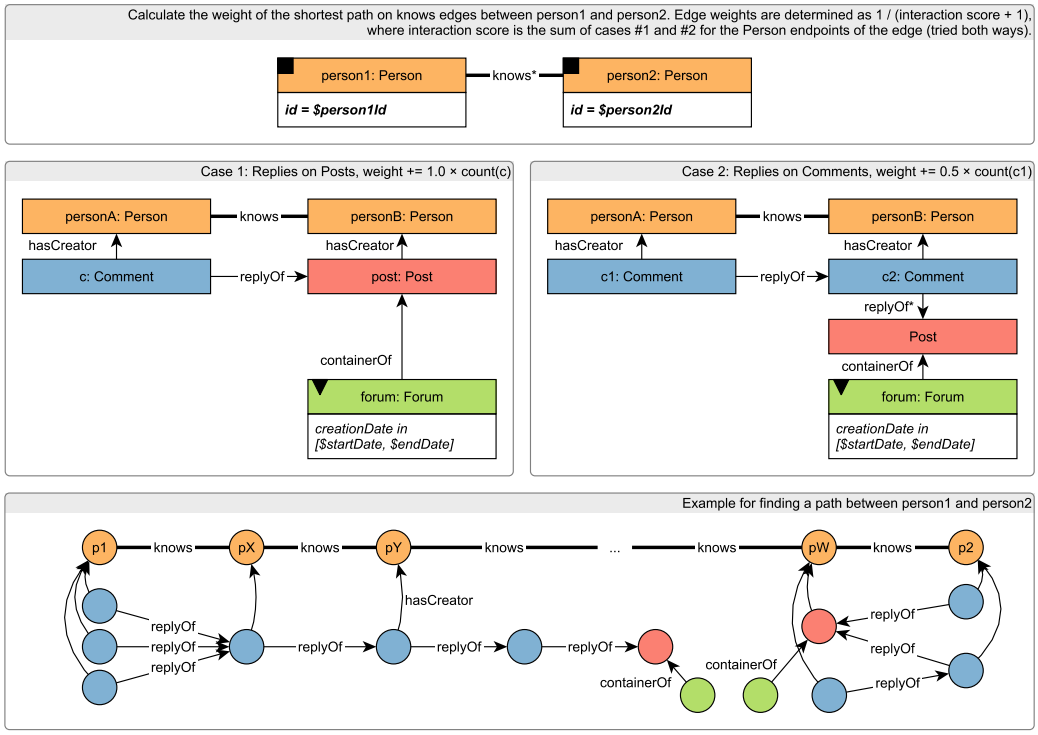
| | | | | |
|---|---|---|---|---|
| 1 | person1.id | ID | R | |
| 2 | person2.id | ID | R | |
| 3 | city1.name | Long String | R | |
| 4 | score | 32-bit Integer | C | |

sort

| | | | |
|---|---|---|---|
| 1 | score | ↓ | |
| 2 | person1.id | ↑ | |
| 3 | person2.id | ↑ | |

| | |
|---|---|
| limit | 100 |
| CPs | 1.3, 1.4, 2.1, 3.1, 3.3, 5.1, 5.2, 5.3, 8.3, 8.4 |

## BI / read / 15

| | |
|---|---|
| query | BI / read / 15 |
| title | Trusted connection paths through forums created in a given timeframe |
| pattern |  |



| | |
|---|---|
| description | Given two Persons with IDs `$person1Id` and `$person2Id`, calculate the cost of the weighted shortest path between these two Persons, in the subgraph induced by the knows relationship. The interaction score of a knows edge is calculated based on the interactions of its Person endpoints:<br>• Every direct reply (by one of the Persons) to a Post (by the other Person) is 1.0 point.<br>• Every direct reply (by one of the Persons) to a Comment (by the other Person) is 0.5 points.<br>Only consider Messages that were created in a Forum that was created within the timeframe (interval) `[$startDate, $endDate]`. Note that for Comments, the containing Forum is that of the Post that the comment (transitively) replies to. Also note that interactions are counted both ways.<br>The weight for the shortest path algorithm is determined as $\frac{1}{interaction\ score+1}$.<br>The result of the query is a single number, the cost of the weighted shortest path. If no such path exists, the query should return −1.0. |

params

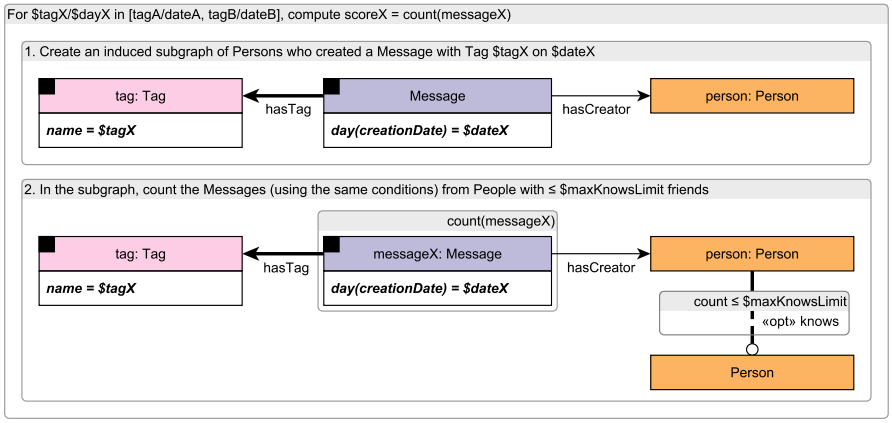
| | | | |
|---|---|---|---|
| 1 | `$person1Id` | ID | (a) `$person1Id` – `$person2Id` pair with a distance of 4 hops<br>(b) `$person1Id` – `$person2Id` pair with a distance of 2 hops |
| 2 | `$person2Id` | ID | |
| 3 | `$startDate` | Date | (a) Small interval (approx. one week)<br>(b) Big interval (approx. one month) |
| 4 | `$endDate` | Date | |

result

| | | | | |
|---|---|---|---|---|
| 1 | `weight` | 32-bit Float | C | |

| | |
|---|---|
| limit | n/a |
| CPs | 1.2, 2.1, 2.2, 2.4, 3.3, 5.1, 5.3, 7.2, 7.3, 7.6, 7.7, 8.1, 8.2, 8.3, 8.4, 8.5, 8.6 |

## BI / read / 16

| | |
|---|---|
| query | BI / read / 16 |
| title | Fake news detection |
| pattern |  |
| description | Given two Tag/date pairs (`$tagA`/`$dateA` and `$tagB`/`$dateB`), for each pair `$tagX`/`$dateX`:<br>• Create an induced subgraph between Persons where for each pair of Persons `person1`/`person2`, both have created a Message on the day of `$dateX` with Tag `$tagX`.<br>• In the induced subgraph, only keep pairs of Persons who have at most `maxKnowsLimit` friends (in the induced subgraph).<br>• For these Persons, count the number of Messages created on `$dateX` with Tag `$tagX`.<br>Return Persons who had at least one Messages for both `$tagA`/`$dateA` and `$tagB`/`$dateB` ranked by their total number of Messages (descending). |

params

| | | | |
|---|---|---|---|
| 1 | `$tagA` | Long String | (a) `$tagA`/`$dateA`, `$tagB`/`$dateB` are both selected to be a flashmob Tag/date combination<br>(b) `$tagA`/`$dateA`, `$tagB`/`$dateB` are both selected to be a non-flashmob Tag/date combination |
| 2 | `$dateA` | Date | |
| 3 | `$tagB` | Long String | |
| 4 | `$dateB` | Date | |
| 5 | `$maxKnowsLimit` | 32-bit Integer | Selected between 3 and 6 |

result

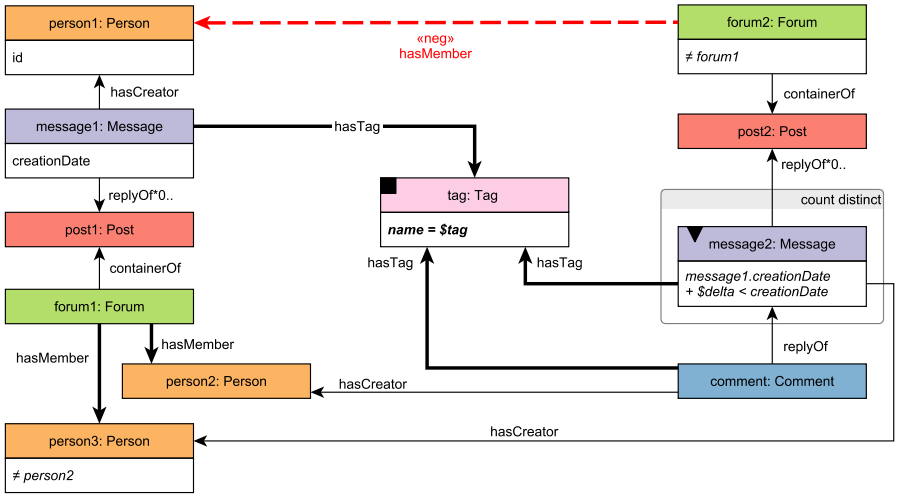
| | | | | |
|---|---|---|---|---|
| 1 | `person.id` | ID | R | |
| 2 | `messageCountA` | 32-bit Integer | A | Message count for `$tagA`/`$dateA` |
| 3 | `messageCountB` | 32-bit Integer | A | Message count for `$tagB`/`$dateB` |

sort

| | | | |
|---|---|---|---|
| 1 | `messageCountA` + `messageCountB` | ↓ | |
| 2 | `person.id` | ↑ | |

| | |
|---|---|
| limit | 20 |
| CPs | 5.3, 8.4, 8.5 |
| relevance | There are two major ways to compute this query: (1) create the induced subgraph as suggested by the specification (either as a view or in materialized form), or (2) skip creating the induced subgraph and perform on-the-fly check for the number of friends (who also posted at least one Message with the given Tag on the given date). The latter approach is easier to express in systems which do not provide graph views but might result in redundant computations (the query engine might repeatedly check whether a Person has at least one Message that satifies the conditions). |

## BI / read / 17

| | |
|---|---|
| query | BI / read / 17 |
| title | Information propagation analysis |
| pattern |  |
| description | This query aims to identify instances of "information propagation" when a Person (`person1`) submits a Message (`message1`) with a given `$tag` to a Forum (`forum1`). This is read by other members of `forum1`, Persons `person2` and `person3` (who must be different Persons). Some time later (specified by the `$delta` parameter), these persons have a discussion with the same `$tag` in a different Forum (`forum2`) where `person1` is not a member. The discussion consists of a Message (`message2`) by `person2` and a direct reply Comment (`comment`) by `person3`.<br>Return IDs of `person1` with the number of interactions their Messages (might have) caused. |



params

| | | | |
|---|---|---|---|
| 1 | $tag | Long String | Tags with a similar amount of Messages are selected |
| 2 | $delta | 32-bit Integer | Measured in hours, selected to be between 8 and 16 hours. |

result

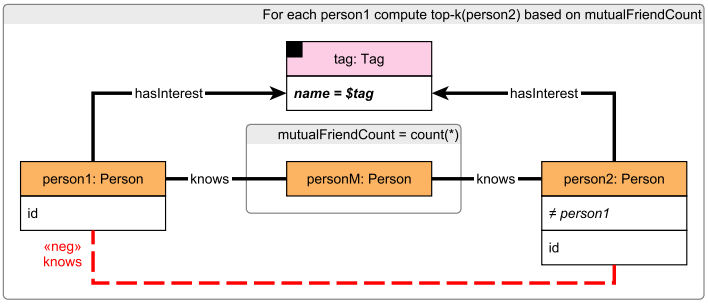
| | | | | |
|---|---|---|---|---|
| 1 | person1.id | ID | R | |
| 2 | messageCount | 32-bit Integer | A | |

sort

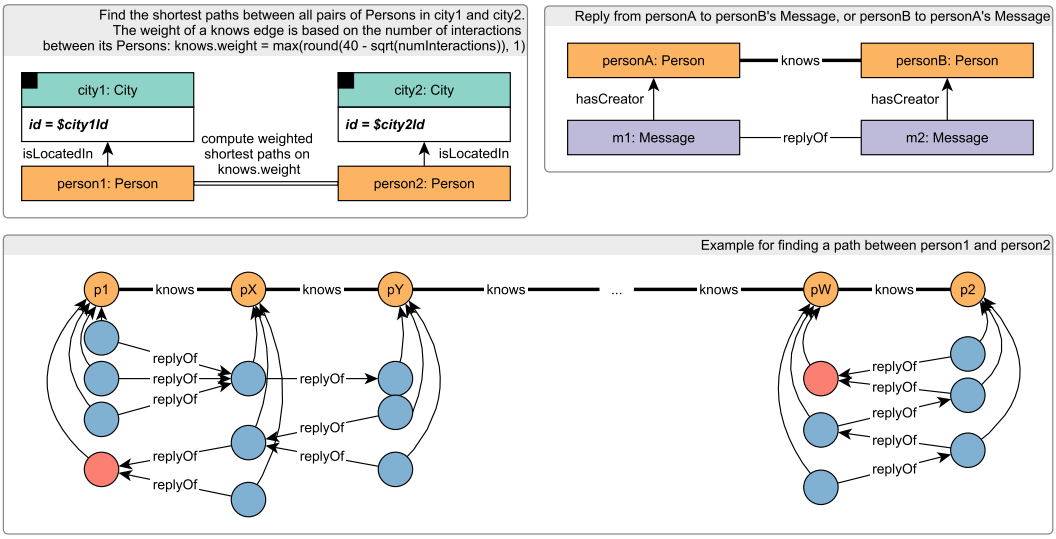
| | | | |
|---|---|---|---|
| 1 | messageCount | ↓ | |
| 2 | person1.id | ↑ | |

| | |
|---|---|
| limit | 10 |
| CPs | 2.1, 2.3, 2.5, 2.6, 8.1 |

## BI / read / 18

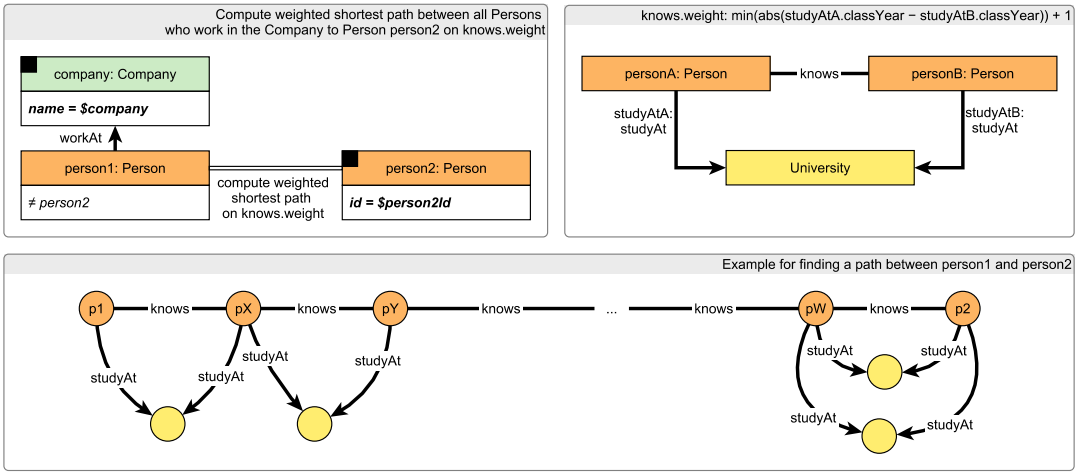
| | |
|---|---|
| query | BI / read / 18 |
| title | Friend recommendation |
| pattern |   |
| description | For a given `$tag`, for each `person1` interested in `$tag`, recommend new friends (`person2`) who<br>• do not yet know `person1`<br>• have at least one mutual friend with `person1`<br>• are also interested in `$tag`.<br>Rank Persons `person2` based on the number of mutual friends with `person1`. |
| params | 1 \| `$tag` \| Long String \| Tags with a similar amount of Persons are selected |
| result | 1 \| `person1.id` \| ID \| R<br>2 \| `person2.id` \| ID \| R<br>3 \| `mutualFriendCount` \| 32-bit Integer \| A |
| sort | 1 \| `mutualFriendCount` \| ↓<br>2 \| `person1.id` \| ↑<br>3 \| `person2.id` \| ↑ |
| limit | 20 |
| CPs | 2.5, 2.6, 8.1 |

## BI / read / 19

| | |
|---|---|
| query | BI / read / 19 |
| title | Interaction path between cities |
| pattern |   |
| description | Given two Cities with IDs `$city1Id`, `$city2Id`, find Persons `person1`, `person2` living in these Cities (respectively) with the *cheapest* interaction path between them.<br>The cheapest path is equivalent to the *weighted shortest* path. It is computed on a subgraph of the Person-knows-Person graph with the edge weights based on the number of interactions. An *interaction* is a direct reply Comments from one Person to Messages by the other Person. Only knows edges with at least one interaction between their endpoint Persons are considered. For these, the weight of a knows edge is defined as: $\max(\mathrm{round}(40 - \sqrt{numInteractions}), 1)$<br>If there are multiple pairs of people with cheapest paths that have the same total weight, return all of them.<br>*Note:* Interactions are counted both ways, e.g. if Alice knows Bob, Alice writes 2 reply Comments to Bob's Messages and Bob writes 3 reply Comments to Alice's Messages, their total number of interactions is 5 and the weight of the knows edge is 38.<br>*Remark:* Determinism is ensured by using square root followed by rounding. For all integers between 1 and 100 000, the square root's fractional part is more than 10e-5 from 0.5, where the rounding could be non-determinstic based on floating point inaccuracies. As 10e-5 is significantly larger than the machine epsilon of IEEE 754 floats (both 32- and 64-bit), the floating point inaccuracies have no chance to affect the derived integer edge weights. |
| params | 1 `$city1Id` ID (a) Small Cities within the same Country (b) Larger Cities from different Countries<br>2 `$city2Id` ID |
| result | 1 `person1.id` ID R<br>2 `person2.id` ID R<br>3 `totalWeight` 32-bit Integer C |
| sort | 1 `person1.id` ↑<br>2 `person2.id` ↑ |
| limit | n/a |
| CPs | 3.3, 7.6, 7.7, 8.4, 8.6 |
| relevance | To find the weighted shortest paths efficiently, the system can use e.g. a bidirectional Dijkstra algorithm. As the edge weights do not depend on any parameter, systems can pre-compute them (if they do not interleave reads and writes). |

## BI / read / 20

BI 1
BI 2
BI 3
BI 4
BI 5
BI 6
BI 7
BI 8
BI 9
BI 10
BI 11
BI 12
BI 13
BI 14
BI 15
BI 16
BI 17
BI 18
BI 19
BI 20

| | |
|---|---|
| query | BI / read / 20 |
| title | Recruitment |
| pattern |  |
| description | Consider knows edges where the endpoint Persons attended the same University and set the weight of the edge to the absolute difference between the year of enrolment plus 1. If the Persons attended multiple universities, we select the smallest (`min`) value. Formally: $$w = \min_{\text{studyAt}_A,\text{studyAt}_B} \lvert \text{studyAt}_A.\text{classYear} - \text{studyAt}_B.\text{classYear} \rvert + 1$$ Given a `$company` and a Person `person2` with ID `$person2Id` (who is not working and has not worked at `$company`), find a different Person (`person1`) who works or at some point worked in `$company` and is reachable from `person2` through people who have studied together through the shortest weighted path.<br>If there are multiple Person `person1` nodes with the same shortest path length, return all of them. |
| params | 1 \| `$company` \| Long String \| Companies with a similar number of employees (former or current) are selected<br>2 \| `$person2Id` \| ID \| (a) There is guaranteed to be no path between any `person1` working at `company` and `person2`<br>(b) There is guaranteed to be a 2-hop path between at least one `person1` working at `company` and `person` |
| result | 1 \| `person1.id` \| ID \| R<br>2 \| `totalWeight` \| 32-bit Integer \| C |
| sort | 1 \| `totalWeight` \| ↑<br>2 \| `person1.id` \| ↑ |
| limit | 20 |
| CPs | 3.3, 7.6, 7.7, 7.8, 8.4, 8.6 |
| relevance | To find the weighted shortest paths efficiently, the system can use e.g. a bidirectional Dijkstra algorithm. As the edge weights do not depend on any parameter, systems can pre-compute them (if they do not interleave reads and writes). |